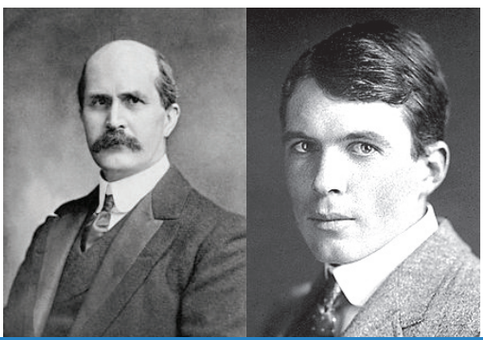
UNIT
6

SOLID STATE



Sir William Henry Bragg (1862 –1942) | Sir Lawrence Bragg (1890 –1971)

Sir William Henry Bragg was a British physicist, chemist, and a mathematician. Sir William Henry Bragg and his son Lawrence Bragg worked on X-rays with much success. They invented the X-ray spectrometer and founded the new science of X-ray crystallography, the analysis of crystal structure using X-ray diffraction. Bragg was joint winner (with his son, Lawrence Bragg) of the Nobel Prize in Physics in 1915, for their services in the "analysis of crystal structure by means of ray". The mineral Braggite (a sulphide ore of platinum, palladium and Nickel) is named after him and his son.

7DMTP

Learning Objectives

After studying this unit, the students will be able to

- describe general characteristics of solids
- distinguish amorphous and crystalline solids
- define unit cell
- describe different types of voids and close packed structures
- calculate the packing efficiency of different types of cubic unit cell
- solve numerical problems involving unit cell dimensions
- explain point defects in solids

XII U6 Solid State - Jerald.indd 176 22-12-2022 19:52:35